

Refining the H1 Crosswalk Frontier: Three Independent Production Roots after Archive Separation and Formal Gluing

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Abstract

TPC-162 records the first missing structural object as the monolithic node H_{mono} . TPC-163–165 now reveal its internal dependency structure without constructing the production carrier. We freeze the TPC-162 pointer as historical fact and refine it by a source-locked acyclic graph.

Inside this monolithic-crosswalk sub-DAG, the graph has exactly three minimal NOT_TESTABLE roots:

H1.source_backed_local_occurrence_edge_family,
H1.actual_active_support_certificate,
H1.canonical_minimal_representation_certificate.

The first root feeds overlap compatibility and formal local-to-global totality. Formal totality then joins the other two roots at a production occurrence witness, which feeds the historical monolithic crosswalk. The exact archived separation key of TPC-164 and the finite gluing theorem of TPC-165 are proved supporting nodes, not unresolved roots.

For work ordering, the local-edge family is selected as the refined first-child pointer. The full three-element antichain is nevertheless preserved: selection is not implication. TPC-163 still finds zero theorem-backed production local occurrence edges, so the verdict remains NOT_TESTABLE. No selected route is stopped, and no actual-carrier impossibility is inferred. These three roots are not the complete H1 blocker set: the full map clause still has nine zero-defect requirements and an independent occurrence registry, while the scalar-plus-ETO clause remains independently NOT_TESTABLE.

Keywords: fixed shift; occurrence provenance; dependency DAG; minimal blocker antichain; route decision.

Claim boundary. This paper refines a frontier; it does not close it. It does not rewrite TPC-162, prove a production local occurrence family, certify actual active support, prove a canonical/minimal actual representation, complete the production crosswalk, stop the selected route, establish positive fixed- X L2, give a prime-pair lower bound, or prove the twin-prime conjecture. The three-root antichain is scoped only to the monolithic-crosswalk sub-DAG, not to the complete H1 contract.

1 Historical pointer versus current refinement

TPC-162 freezes the current verdict

$$\text{Verdict}_{162} = \text{NOT-TESTABLE}$$

and the first missing object

$$H_{\text{mono}} := H1.\text{theorem_backed_occurrence_provenance_crosswalk}. \quad (1.1)$$

This statement is correct at the interface resolution used there [2]. A later decomposition should not retroactively replace (1.1): reproducible route histories require old pointers to remain interpretable against their own source locks.

TPC-163–165 provide new resolution:

1. the frozen source corpus has no theorem-backed production actual-occurrence edge, and native triples collide [3];
2. a unique five-field minimum key separates every archived cut row [4]; and
3. compatible local formal patches would glue uniquely with exact conservation, but no production patch is supplied [5].

These facts justify a new child-level graph beneath H_{mono} .

1.1 The complete H1 envelope remains outside this refinement

TPC-156 defines two sufficient clauses [1]. Write

$$\mathcal{D} = \{D_L, D_{QD}, D_{QZ}, D_G, D_P, D_{DZ}, D_{GP}, D_{\text{cover}}, D_{\text{rec}}\}.$$

Then

$$\begin{aligned} \text{Map}_{H1} &= H_{\text{mono}} \wedge \bigwedge_{D \in \mathcal{D}} (D = 0) \wedge \text{OccurrenceRegistryTotal}, \\ \text{Scalar}_{H1} &= \text{CompleteFUM}_{o(X)} \wedge \text{TheoremBackedETO}, \\ \text{H1Complete} &= \text{Map}_{H1} \vee \text{Scalar}_{H1}. \end{aligned} \tag{1.2}$$

TPC-166 refines only the factor H_{mono} in Map_{H1} . The nine zero-defect conditions and the independent occurrence registry remain external requirements. The scalar-plus-ETO clause is a separate NOT-TESTABLE route and is not a child of H_{mono} . Therefore the antichain computed below must never be reported as the complete H1 minimal blocker antichain.

Definition 1.1 (Non-retroactive refinement). A refinement of a historical frontier node H consists of:

- (i) a source lock preserving H 's original identifier, status, and scope;
- (ii) a directed acyclic graph whose target is a new copy of H ;
- (iii) typed child nodes whose conjunction supplies H ; and
- (iv) an explicit statement that the historical record is not rewritten.

2 The refined production DAG

We abbreviate:

$$\begin{aligned} H_{\text{local}} &= \text{source-backed local occurrence edge family,} \\ H_{\text{support}} &= \text{actual active support certificate,} \\ H_{\text{canon}} &= \text{canonical/minimal representation certificate,} \\ H_{\text{overlap}} &= \text{local overlap bijection/cocycle,} \\ H_{\text{total}} &= \text{glued formal occurrence totality,} \\ H_{\text{witness}} &= \text{production occurrence witness.} \end{aligned}$$

The dependency direction is “parent is prerequisite.” The full refined graph is:

$$\begin{array}{ll} H_{\text{local}} & \longrightarrow H_{\text{overlap}} \\ \{H_{\text{local}}, H_{\text{overlap}}, K_{164}, G_{165}\} & \longrightarrow H_{\text{total}} \\ \{H_{\text{total}}, H_{\text{support}}, H_{\text{canon}}\} & \longrightarrow H_{\text{witness}} \\ H_{\text{witness}} & \longrightarrow H_{\text{mono}}, \end{array} \tag{2.1}$$

where K_{164} is the proved archived separation key and G_{165} is the proved finite typed gluing theorem. The three nodes $H_{\text{local}}, H_{\text{support}}, H_{\text{canon}}$ have no parents. The nodes K_{164}, G_{165} are also parentless, but their status is PROVED, so they are not blockers.

Definition 2.1 (Minimal unresolved root antichain). For a target H in a finite prerequisite DAG, take the ancestral subgraph of H . An unresolved root is a node of status NOT-TESTABLE having no NOT-TESTABLE parent in that subgraph. The set of all such roots is the *minimal unresolved root antichain*.

This definition removes descendants that are unavailable only because an earlier unresolved prerequisite is missing. It does not choose one root at the expense of independent roots.

3 Exact three-root frontier

Theorem 3.1 (Refined crosswalk-sub-DAG blocker antichain). *Within the source-locked DAG (2.1), the minimal unresolved root antichain of H_{mono} is exactly*

$$\mathcal{A}_{166} = \{H_{\text{local}}, H_{\text{support}}, H_{\text{canon}}\}. \quad (3.1)$$

The nodes $H_{\text{overlap}}, H_{\text{total}}, H_{\text{witness}}, H_{\text{mono}}$ are unresolved descendants and hence do not belong to $\mathcal{A}_{166}$.

Proof. The imported TPC-164 and TPC-165 audits give

$$\text{status}(K_{164}) = \text{status}(G_{165}) = \text{PROVED}.$$

TPC-163 gives zero qualifying production local edges, and TPC-165 keeps formal totality, actual support, and canonical/minimality separate. Therefore

$$\text{status}(H_{\text{local}}) = \text{status}(H_{\text{support}}) = \text{status}(H_{\text{canon}}) = \text{NOT-TESTABLE}.$$

Each of these three has no unresolved parent, so all belong to the root antichain.

The node H_{overlap} has unresolved parent H_{local} . The node H_{total} has unresolved parents $H_{\text{local}}, H_{\text{overlap}}$. The node H_{witness} has unresolved parents $H_{\text{total}}, H_{\text{support}}, H_{\text{canon}}$, and H_{mono} has unresolved parent H_{witness} . Hence none of these descendants is minimal. The DAG contains no other nodes, proving (3.1). $\square$

Remark 3.2 (Not the full H1 antichain). [Theorem 3.1](#) concerns only the ancestral sub-DAG of H_{mono} . Replacing H_{mono} by its three roots inside (1.2) still leaves the nine defect nodes, occurrence-registry totality, and the independent scalar-plus-ETO clause. Their full-DAG minimality is not computed here.

Corollary 3.3 (No single-root reduction). *Proving any one member of $\mathcal{A}_{166}$ does not, by the current dependency contract, prove either of the other two.*

Proof. The three nodes are pairwise parent-independent in (2.1). No imported theorem supplies a directed path between them. $\square$

4 A selected pointer that preserves the antichain

For sequential work, a single display pointer is useful. We select

$$\text{FirstChild}_{166} = H_{\text{local}} \quad (4.1)$$

because it is the first source-producing object on the local-to-global branch. If local edges are found, overlap maps become testable and the proved gluing theorem can be instantiated.

Proposition 4.1 (Selection is not erasure). *Equation (4.1) is an ordering convention for the next construction. The production blocker set remains the full $\mathcal{A}_{166}$ of [equation \(3.1\)](#).*

Proof. The machine decision stores both fields independently:

$$\begin{aligned} \text{canonical selected representative} &= H_{\text{local}}, \\ \text{full antichain preserved} &= [H_{\text{local}}, H_{\text{support}}, H_{\text{canon}}]. \end{aligned}$$

Its validator recomputes the antichain from the DAG and rejects any candidate that deletes H_{support} or H_{canon} . Thus selection cannot change the mathematical dependency set. $\square$

Here “canonical selected representative” refers only to deterministic serialization and display order. It is not the missing H_{canon} theorem.

5 Production evidence and verdict

The source-locator census remains decisive at the first child:

$$\#\mathcal{E}_{\text{local}}^{\text{theorem-backed}} = 0 \quad \text{in the frozen declared corpus.} \quad (5.1)$$

Consequently no production local patch family exists in the current evidence set. The statuses are:

Object	Status
minimum archived separation key	PROVED
finite typed gluing theorem	PROVED
source-backed local occurrence edge family	NOT-TESTABLE
actual active-support certificate	NOT-TESTABLE
canonical/minimal representation certificate	NOT-TESTABLE
overlap cocycle and glued formal totality	NOT-TESTABLE
production occurrence witness and monolithic crosswalk	NOT-TESTABLE

Theorem 5.1 (Current refined decision). *Under the nine source locks of TPC-166, and only inside the historical monolithic-crosswalk sub-DAG,*

$$\boxed{\text{Verdict}_{166} = \text{NOT-TESTABLE}}, \quad \text{FirstChild}_{166} = H_{\text{local}}, \quad \mathcal{A}_{166} = \{H_{\text{local}}, H_{\text{support}}, H_{\text{canon}}\}. \quad (5.2)$$

The selected occurrence-augmented route is not stopped, and no actual-carrier impossibility is proved.

Proof. The verdict cannot be GO because each root in $\mathcal{A}_{166}$ is unresolved. It cannot be a route stop or impossibility verdict because the source census is closed-world and the formal gluing theorem leaves source augmentation open. Thus the typed result is NOT-TESTABLE, with pointer and antichain given by [equations \(4.1\)](#) and [\(3.1\)](#). $\square$

6 Why TPC-162 is not rewritten

There is no contradiction between

$$\text{FirstMissing}_{162} = H_{\text{mono}} \quad \text{and} \quad \text{FirstChild}_{166} = H_{\text{local}}.$$

The first names the unresolved public interface at TPC-162 resolution. The second names one child after a later factorization. The certificate records

$$\text{historical pointer retroactively rewritten} = \text{false}$$

and mutation-tests the opposite value. This provenance discipline allows later papers to refine a route without silently changing what an earlier source actually asserted.

7 Reproducible DAG audit

The generator locks the TPC-162 snapshot/audit and both certificate/audit pairs from TPC-163–165, together with the TPC-156 full H1 contract. It validates:

1. unique node identifiers and total parent references;
2. acyclicity by depth-first topological traversal;
3. exact node statuses and dependency lists;
4. recomputation of the minimal unresolved root antichain;
5. preservation of the crosswalk-sub-DAG scope and exclusion of the nine defects, independent registry, and scalar clause;
6. preservation of the historical pointer;
7. zero production local edges and the NOT-TESTABLE verdict.

Real mutations delete either independent root, rewrite the historical pointer, fabricate a production edge, promote the verdict to GO, stop the route, introduce a dependency cycle, or select a representative outside the antichain. The same validator rejects each mutation.

8 Next source-producing task

The refinement identifies what the next paper must actually deliver: not another global schema, but at least one theorem-backed local actual-occurrence edge family with:

1. the five-field archived source address;
2. canonical path and hash, theorem and formula locators, and a nonempty derivation tree;
3. exact typed row weights and parent/stage/downstream payloads;
4. an explicit statement of whether the result is merely formal or lies on the actual carrier.

A partial local theorem would be genuine forward progress even before total coverage. It would not close H_{support} or H_{canon} ; those remain parallel research programs.

9 Conclusion

TPC-166 turns one opaque H1 blocker into a precise three-root frontier inside that blocker. Two supporting uncertainties are already closed: archived cuts have an exact lossless key, and compatible formal patches glue with exact conservation. The unresolved mathematics is now localized to source-backed local occurrence edges, actual active support, and canonical/minimal representation. The route remains open, the verdict remains NOT-TESTABLE, and the next constructive target is smaller without being overstated. The complete H1 graph, including the external map-clause and scalar-clause requirements of (1.2), remains for a later integration.

References

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